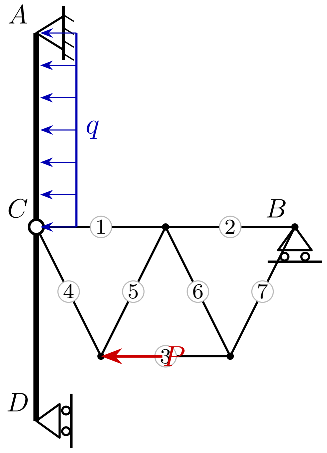
University of Natural Resources and Life Sciences, Vienna (BOKU)

VU Mechanik – LAWI 100515 (PI)

Exam **Group A** Date: 26.06.2026

Given system

The planar system consists of bending beams and a truss connected by hinges.



Given quantities: $a = 3\,\mathrm{m}$ $\ell = 2\,\mathrm{m}$ $h = 2\,\mathrm{m}$ $q = 4\,\mathrm{kN/m}$ $P = 10\,\mathrm{kN}$

Tasks

1. Determine the degree of static determinacy.
2. Compute all support reactions and hinge forces.
3. Determine all truss member forces (tension/compression).
4. Draw the section forces N, V, M of the bending beams.